LIMITLESS MODUS

AI-NATIVE ENTERPRISE TRANSFORMATION

Self-submission Audit Wizard Specification

Invincibility Blueprint Toolkit

Complete specification for the Self-submission Audit Wizard — defining step flows, question types, conditional routing, data model, authentication, API contract, and Nano Fibre UK engagement configuration. Blueprint for implementation in limitless-portal.

Self-submission Audit Wizard — Specification

> ***Version:*** *1.0 (2 Mar 2026)* ***Status:*** *Draft — for review before implementation* ***Target repository:*** *tumai-programmes/limitless-portal* ***First engagement:*** *Nano Fibre UK (27 participants, audit go-live 10 Mar 2026)*

Contents

1. Overview and Design Principles
2. Wizard Step Flow — 01 Company Audit
3. Wizard Step Flow — 02 Manager Audit
4. Wizard Step Flow — 03 Engineer Mini-Audit
5. Conditional Routing Rules
6. Data Model
7. Authentication and Authorisation
8. Participant Journey
9. Admin Dashboard Requirements
10. File Upload and Evidence Management
11. Auto-save, Progress Tracking, and Submission Lifecycle
12. API Contract Outline
13. Appendix A — Nano Fibre UK Engagement Configuration

1. Overview and Design Principles

1.1 What This Specifies

This document defines the interactive web-based wizard that delivers the three Invincibility Blueprint self-submission audit instruments:

#	Instrument	Audience	Wizard Steps	Time	Conditional Logic
01	Company Audit	Director / MD	13 (Consent + 1 + A–K)	60–90 min	None — all sections required
02	Manager Audit	Functional managers	15 (Consent + 1–13)	30–45 min	Section 5 routes by department
03	Engineer Mini-Audit	Field engineers	12 (Consent + 1–10)	10–12 min	Section 5 routes by job type

The wizard replaces Greg's current Word docs + Microsoft Forms delivery with an interactive, auto-saving, section-by-section web experience.

1.2 Design Principles

1. **Speed and truth, not presentation** — the UI must make it faster to submit than to format a Word document. Pre-populated fields, clear skip paths, and auto-save reduce friction.
2. **Section-at-a-time** — each wizard step maps to one section of the instrument. Participants see progress and can return to any completed section.
3. **ResponseMode pattern** — for every evidence/upload request, offer the four response modes: Upload / Provide link / Not available (covered in interviews) / Will send later (ETA).
4. **Mobile-aware** — field engineers will likely complete the Mini-Audit on phones or tablets. The wizard must be responsive.
5. **No blame, evidence over polish** — the UI reinforces this message at every step. No red validation errors for incomplete sections — just a "complete" vs "in progress" indicator.
6. **Draft persistence** — every field auto-saves. Participants can close and resume at any point.
7. **Confidentiality first** — consent gate before any data entry. Upload guidance shown before every file input.

1.3 Question Type Taxonomy

Every question across all three instruments maps to one of these input types:

Type	Widget	Validation	Example

`FreeText`	Single-line `<input>` or multi-line `<textarea>`	Optional min/max length	"Legal entity name + trading name"
`FreeTextLong`	Multi-line `<textarea>` with rich-text option	Optional min length	"Describe your operational system"
`SingleSelect`	Radio buttons or dropdown	Must select one	OD Stage self-assessment (1–4)
`MultiSelect`	Checkbox group	Optional min/max selections	"Primary services delivered (tick all)"
`MultiSelectCapped`	Checkbox group with max constraint	Max N selections	"Pick top 2 categories"
`FileUpload`	Drag-and-drop file zone + metadata	Max file size, allowed types	"Upload org chart"
`ResponseMode`	4-option radio group + conditional sub-fields	Must select one mode	"Upload what we have / Provide link / Not available / Will send later"
`TableGrid`	Structured table with editable cells	At least 1 row	Non-productive time estimates, Regular Duties cadence
`RatingScale`	Dropdown or radio per item (8-point scale)	Must rate each standard item	Reality Test ratings
`RatingScaleWithEvidence`	Rating dropdown + FreeText + optional FileUpload per item	Rating required, evidence optional	Reality Test with evidence description
`ConsentCheckbox`	Single checkbox	Must be checked to proceed	GDPR consent
`NumberInput`	Numeric input with optional unit	Optional range	"How many disciplinaries are open?"
`ChecklistUpload`	Checkbox list where each item has a ResponseMode sub-field	At least awareness (check or skip)	B2.2 department-by-department uploads

2. Wizard Step Flow — 01 Company Audit

Audience: Director / MD / CEO **Total steps:** 13 (Consent + Company Context + Sections A–K) **Estimated time:** 60–90 minutes **Conditional logic:** None — all sections required

Step 0: Consent

#	Question	Type	Required
0.1	Confidentiality statement (read-only display)	—	Display only
0.2	Upload guidance (read-only display)	—	Display only
0.3	"Before You Start" guidance (read-only display)	—	Display only
0.4	Consent checkbox: "I confirm I am authorised..."	`ConsentCheckbox`	Yes — gate

Gate: Cannot proceed to Step 1 until consent is checked.

Step 1: Company Context

#	Question	Type	Required
1.1	Legal entity name + trading name	`FreeText`	Yes
1.2	Primary services delivered	`MultiSelect`	Yes
	Options: Broadband installations / Service calls / Pre-Enablement / Enablement works		
1.3	Clients: network operators/ISPs you deliver for	`FreeTextLong`	Yes
1.4	Geographic footprint: regions, depots/stores coverage	`FreeTextLong`	Yes
1.5	Headcount & workforce mix (FTE vs contractors, by function)	`FreeTextLong`	Yes
1.6	Volume profile (last 12 weeks)	`TableGrid`	Yes
	Columns: Workstream / Volume per week		
	Rows: Installs / Service calls / Pre-enablement / Enablement		
1.7	Top 5 priorities (next 12 months)	`FreeTextLong`	Yes
1.8	Company overview pack	`ResponseMode`	No
1.9	Catch-all: what else exists (Context)	`FreeTextLong` + `FileUpload`	No

Step 2: Section A — Organisational Foundation

#	Question	Type	Required
A1.1	Organisation purpose statement (1 paragraph)	`FreeTextLong`	Yes
A1.2	Core values + delivery behaviours	`FreeTextLong`	Yes
A1.3	Non-negotiables (behaviours not tolerated)	`FreeTextLong`	Yes
A1.4	Purpose/values documents	`ResponseMode`	No
A2.1	Operational system description	`FreeTextLong`	Yes
A2.2	3–5 stability rules	`FreeTextLong`	Yes
A2.3	Operating model/playbook	`ResponseMode`	No
A3.1	Org chart / organogram	`ResponseMode`	Yes
A3.2	Top-level functions list	`FreeTextLong`	Yes
A4.1	Financial approvals thresholds	`FreeTextLong`	Yes
A4.2	Resourcing approvals	`FreeTextLong`	Yes
A4.3	Hiring/firing authority	`FreeTextLong`	Yes
A4.4	Contract/commercial approvals	`FreeTextLong`	Yes
A4.5	HSEQ stop-work authority	`FreeTextLong`	Yes
A4.6	Escalation thresholds	`FreeTextLong`	Yes
A4.7	DoA matrix / decision schedule	`ResponseMode`	Yes
A.catch	Catch-all (Foundation)	`FreeTextLong` + `FileUpload`	No

Step 3: Section B — Organisation Structure, Departments & Roles

#	Question	Type	Required
B1.1	Latest organisation chart	`ResponseMode`	Yes
B1.2	Departments/teams list (purpose, key outputs, interfaces)	`FreeTextLong`	Yes
B1.3	Team charters / department summaries	`ResponseMode`	No
B2.1	Regular Duties guidance (read-only display)	—	Display only
B2.2	Department-by-department uploads	`ChecklistUpload`	Yes
	Items: Installs / Service / Pre-Enablement / Enablement / Dispatch / Stores / QA / HSEQ / Client Performance / Support functions / Custom		
	Each item: `ResponseMode` (Upload / Link / Not available / Will send later)		
B3.1	Role descriptions / job descriptions	`ResponseMode`	Yes
B3.2	Per management layer: role purpose, top 5 outcomes, key decisions	`FreeTextLong`	Yes
B4.1	Critical Outcomes Ownership (10 items)	`TableGrid`	Preferred
	Columns: Outcome / Owner / How measured / Current performance		
	Rows: 10 pre-populated outcomes		
B.catch	Catch-all (Roles)	`FreeTextLong` + `FileUpload`	No

Step 4: Section C — Workforce, Capability, Performance & Discipline

#	Question	Type	Required
C1.1	Headcount by department/function	`FreeTextLong`	Yes
C1.2	Attrition last 12 months + top 3 reasons	`FreeTextLong`	Yes
C1.3	Contractor reliance %	`FreeTextLong`	Yes
C1.4	Anonymised headcount report	`ResponseMode`	Preferred
C2.1	Mandatory competencies/accreditations	`FreeTextLong`	Yes
C2.2	How competence is tracked/refreshed/audited	`FreeTextLong`	Yes
C2.3	Competence matrix / training records	`ResponseMode`	Preferred
C3.1	Performance measurement approach	`FreeTextLong`	Yes
C3.2	Last 3 months performance reporting	`ResponseMode`	Yes
C4.1	Open disciplinaries breakdown	`FreeTextLong`	Yes
C4.2	Top 3 disciplinary themes	`FreeTextLong`	Yes
C4.3	Time-to-close and where disciplinaries stall	`FreeTextLong`	Yes
C4.4	Anonymised disciplinary summary	`ResponseMode`	Preferred
C5.1	Succession plan for critical roles	`FreeTextLong`	No
C5.2	Where capability breaks if you lose 2 key people	`FreeTextLong`	No
C5.3	Succession plan upload	`ResponseMode`	No
C6.1	Bonus/reward scheme description	`FreeTextLong`	No
C6.2	Where incentives drive wrong behaviours	`FreeTextLong`	No
C.catch	Catch-all (People)	`FreeTextLong` + `FileUpload`	No

Step 5: Section D — Strategy, Planning & Execution Discipline

#	Question	Type	Required
D1.1	Current strategy description	`FreeTextLong`	Yes
D1.2	Top 3 constraints blocking strategy	`FreeTextLong`	Yes
D1.3	Strategy deck / business plan	`ResponseMode`	No
D2.1	Top company goals (3–10) with success criteria, owner, status, dates	`FreeTextLong`	Yes
D2.2	Key objectives/projects per goal	`FreeTextLong`	Yes
D2.3	Top 3 active objectives task plan	`FreeTextLong`	Yes
D2.4	Top 10 overdue tasks/actions and why	`FreeTextLong`	Yes
D2.5	Where plans typically fail	`FreeTextLong`	Yes
D2.6	Goals/project/task trackers	`ResponseMode`	Preferred
D3.1	OD Stage self-assessment	`SingleSelect`	Yes
	Options: Stage 1 (Start-up) / Stage 2 (Regular Management) / Stage 3 (Viral Expansion) / Stage 4 (Synergistic Brotherhood)		
D3.2	Evidence for selected stage (3–5 examples)	`FreeTextLong`	Yes
D3.3	Target stage in 12–24 months and why	`FreeTextLong`	Yes
D.catch	Catch-all (Strategy/Plans)	`FreeTextLong` + `FileUpload`	No

Step 6: Section E — Process, Compliance & Reality Tests

#	Question	Type	Required
E1.1	Process map: Installs	`FreeTextLong` + `ResponseMode`	Yes
E1.2	Process map: Service	`FreeTextLong` + `ResponseMode`	Yes
E1.3	Process map: Pre-Enablement	`FreeTextLong` + `ResponseMode`	Yes
E1.4	Process map: Enablement	`FreeTextLong` + `ResponseMode`	Yes
E1.5	Top 10 break points	`FreeTextLong`	Yes
E1.6	Process maps / SOP index / work instructions	`ResponseMode`	Yes
E2.1	Reality Test: 10 standard components	`RatingScaleWithEvidence`	Yes
	Rating options (8-point): Exists and is used / Exists but not used consistently / Claimed to exist but cannot be found / Exists but under-resourced / Implemented but no effect / Tool used for wrong purpose / Multiple approaches conflict / Works only because specific people carry it		
	Items: Planning & scheduling / Dispatch & job allocation / QA gates / Repeat faults loop / Subcontractor governance / Stores control / HSEQ controls / KPI reporting / FSM-portal integration / Portal closure rules		
E2.2	Additional components (up to 8)	`RatingScaleWithEvidence`	Yes
E3.1	Contract register summary	`ResponseMode`	Yes
E3.2	KPI/SLA schedules and scorecards	`ResponseMode`	Yes
E3.3	Commercial governance description	`FreeTextLong`	Yes
E3.4	Contract extracts (redacted)	`ResponseMode`	No
E.catch	Catch-all (Process/Compliance)	`FreeTextLong` + `FileUpload`	No

Step 7: Section F — Operating Cadence & Control Schedules

#	Question	Type	Required
F.grid	11 schedules (F1–F11)	`ChecklistUpload`	Yes
	Each item: Schedule name + `ResponseMode` (Upload / None + explain)		
	F1: Reporting & communications / F2: Decision-making / F3: KPI / F4: SLA / F5: Absence cover / F6: Escalation / F7: Out-of-hours / F8: Systems / F9: Reports / F10: Documentation / F11: Customer feedback		
F.catch	Catch-all (Cadence/Schedules)	`FreeTextLong` + `FileUpload`	No

Step 8: Section G — External System

#	Question	Type	Required
G.1	Supplier/subcontractor register	`FreeTextLong` + `ResponseMode`	Yes
G.2	Client register	`FreeTextLong` + `ResponseMode`	Yes
G.3	Service catalogue	`FreeTextLong` + `ResponseMode`	Yes
G.4	Top 5 market/industry forces	`FreeTextLong`	Yes
G.catch	Catch-all (External)	`FreeTextLong` + `FileUpload`	No

Step 9: Section H — Finance & Full Cost Audit

#	Question	Type	Required
H1.1	Last 12 months monthly actuals P&L	`ResponseMode`	Yes
H1.2	Last FY P&L + current forecast	`ResponseMode`	Yes
H1.3	Chart of accounts	`ResponseMode`	No
H2.1	Fleet costs	`FreeTextLong`	Yes
H2.2	Tools/test equipment	`FreeTextLong`	Yes
H2.3	Subcontractor costs	`FreeTextLong`	Yes
H2.4	Overtime/standby/on-call	`FreeTextLong`	Yes
H2.5	Rework cost indicators	`FreeTextLong`	Yes
H2.6	Cost driver uploads	`ResponseMode`	No
H3.1	Estates list (purpose, utilisation, cost, lease terms)	`FreeTextLong`	Yes
H3.2	Leases / facilities cost schedule	`ResponseMode`	No
H4.1	Where costs are leaking	`FreeTextLong`	Yes
H4.2	What costs feel normal but are waste	`FreeTextLong`	Yes
H4.3	Where penalties/credits/chargebacks occur	`FreeTextLong`	Yes
H.catch.1	Catch-all (Finance/Costs)	`FreeTextLong` + `FileUpload`	No
H5.1	Inventory value split (warehouse/depots/van/in-transit)	`FreeTextLong`	Yes
H5.2	Inventory ageing by value	`FreeTextLong`	Yes
H5.3	Slow-moving/obsolete % and write-off policy	`FreeTextLong`	Yes
H5.4	Stock turns / Days on Hand	`FreeTextLong`	Yes
H5.5	Stock accuracy (cycle counts vs book)	`FreeTextLong`	Yes
H5.6	Stuck returns value and management	`FreeTextLong`	Yes
H5.7	Client-owned/consignment stock rules	`FreeTextLong`	Yes
H5.uploads	Inventory uploads (7 items)	`ChecklistUpload`	Preferred
H6.1	Unit-rate schedules / rate cards	`ResponseMode`	Yes
H6.2	"Done for payment" description per client	`FreeTextLong`	Yes
H6.3	Throughput pipeline (last 3 months)	`FreeTextLong` + `ResponseMode`	Yes
H6.4	WIP/unbilled backlog	`FreeTextLong`	Yes
H6.5	Top 10 rejected closure reasons	`FreeTextLong`	Yes
H6.6	Top 10 invoice dispute reasons	`FreeTextLong`	Yes
H6.7	Credits/penalties/chargebacks summary	`FreeTextLong`	Yes
H6.uploads	Throughput uploads (5 items)	`ChecklistUpload`	Preferred

Step 10: Section I — Operations Deep Dive

#	Question	Type	Required
I1.1	Install flow description	`FreeTextLong`	Yes
I1.2	Top 10 install failure reasons	`FreeTextLong`	Yes
I1.3	Right First Time % and measurement	`FreeTextLong`	Yes
I1.4	Early Life Failure % and measurement	`FreeTextLong`	Yes
I1.5	Appointment promise control	`FreeTextLong`	Yes
I1.6	Where installs get blocked	`FreeTextLong`	Yes
I1.7	Install uploads	`ResponseMode`	No
I2.1	Service flow description	`FreeTextLong`	Yes
I2.2	SLA targets vs actual	`FreeTextLong`	Yes
I2.3	Repeat fault rate and causes	`FreeTextLong`	Yes
I2.4	Root cause process	`FreeTextLong`	Yes
I2.5	Service uploads	`ResponseMode`	No
I3.1	Pre-enablement flow description	`FreeTextLong`	Yes
I3.2	Common outcomes	`FreeTextLong`	Yes
I3.3	Success measurement	`FreeTextLong`	Yes
I3.4	Where pre-enablement creates waste	`FreeTextLong`	Yes
I3.5	Pre-enablement uploads	`ResponseMode`	No
I4.1	Enablement workflow description	`FreeTextLong`	Yes
I4.2	% installs/service blocked by enablement	`FreeTextLong`	Yes
I4.3	Common enablement types	`FreeTextLong`	Yes
I4.4	Where delays occur	`FreeTextLong`	Yes
I4.5	Enablement uploads	`ResponseMode`	No
I5.1	Dispatch rules	`FreeTextLong`	Yes
I5.2	Non-productive time breakdown	`TableGrid`	Yes
	Columns: Category / Estimate		
	Rows: Van-travel / Stores collection / Waiting access / Waiting records / Waiting permits / Rework / Admin / Portal admin / FSM mismatch / Other		
I5.3	Top 5 causes of lost time per week	`FreeTextLong`	Yes
I5.4	Dispatch uploads	`ResponseMode`	No
I6.1	Materials flow description	`FreeTextLong`	Yes
I6.2	Where losses/waste occur	`FreeTextLong`	Yes
I6.3	Stockout frequency and impact	`FreeTextLong`	Yes
I6.4	Top 20 SKUs by value	`FreeTextLong`	No
I6.5	"Just in case" stock	`FreeTextLong`	Yes
I6.6	Unusable stock value	`FreeTextLong`	No
I6.7	Stores uploads	`ResponseMode`	No
I6B.1	Hold free-issue stock?	`SingleSelect` (Yes/No)	Yes
I6B.2	Which clients and categories	`FreeTextLong`	Conditional (Yes)
I6B.3	Where held	`FreeTextLong`	Conditional (Yes)
I6B.4	How tracked separately	`FreeTextLong`	Conditional (Yes)
I6B.5	Reconciliation method	`FreeTextLong`	Conditional (Yes)
I6B.6	Discrepancy handling	`FreeTextLong`	Conditional (Yes)
I6B.7	Current discrepancy exposure	`FreeTextLong`	Conditional (Yes)
I6B.uploads	Free-issue uploads (5 items)	`ChecklistUpload`	No
I7.1	QA gates description	`FreeTextLong`	Yes
I7.2	How findings become corrective action	`FreeTextLong`	Yes
I7.3	QA uploads	`ResponseMode`	No
I8.1	Safety standards: embedded, audited, enforced	`FreeTextLong`	Yes
I8.2	Last 6–12 months safety summary	`FreeTextLong`	Yes
I8.3	HSEQ uploads	`ResponseMode`	No
I.catch	Catch-all (Operations)	`FreeTextLong` + `FileUpload`	No

Step 11: Section J — Leadership System Signals

#	Question	Type	Required
J.1	% leadership time firefighting vs prevention	`FreeTextLong`	Yes
J.2	Last 5 repeat problems that keep returning	`FreeTextLong`	Yes
J.3	Where important information arrives late	`FreeTextLong`	Yes
J.4	Where senior leaders bypass managers	`FreeTextLong`	Yes
J.5	Where trust in management system is low	`FreeTextLong`	Yes
J.catch	Catch-all (Leadership)	`FreeTextLong` + `FileUpload`	No

Step 12: Section K — Evidence Register

#	Question	Type	Required
K.1	Document register / index	`ResponseMode`	No
K.2	Document list (if no register)	`TableGrid`	No
	Columns: Name / Owner / Location-link / Last updated / What it proves		
K.3	Evidence pack upload	`FileUpload`	No

3. Wizard Step Flow — 02 Manager Audit

Audience: Functional managers (all departments) **Total steps:** 15 (Consent + Sections 1–13) **Estimated time:** 30–45 minutes **Conditional logic:** Section 5 branches by department (selected in Section 1)

Step 0: Consent

#	Question	Type	Required
0.1	Confidentiality statement + upload guidance + "How to Use" (read-only)	—	Display
0.2	Consent checkbox	`ConsentCheckbox`	Yes — gate

Step 1: Section 1 — About You

#	Question	Type	Required
1.1	Name	`FreeText`	Yes
1.2	Role title	`FreeText`	Yes
1.3	Department/team	`SingleSelect`	Yes
	Options: Field Ops – Installations / Field Ops – Service Calls/Repair / Field Ops – Pre-Enablement / Field Ops – Enablement Works / Dispatch/Scheduling / Stores/Materials / QA/Quality / HSEQ / HR / Other		
1.4	Other department (describe)	`FreeText`	Conditional (Other)
1.5	Location/region covered	`FreeText`	Yes
1.6	Who do you report to?	`FreeText`	Yes
1.7	Team size (direct + indirect)	`FreeText`	Yes
1.8	Workforce mix	`SingleSelect`	No
	Options: Employed only / Subcontractors only / Both		

Routing trigger: The value of question 1.3 determines which Section 5 sub-section(s) the participant sees.

Step 2: Section 2 — Role Purpose & What "Good" Looks Like

#	Question	Type	Required
2.1	Purpose of your role (1 paragraph)	`FreeTextLong`	Yes
2.2	Top 5 outcomes you are accountable for	`FreeTextLong`	Yes
2.3	What "good" looks like (targets/standards/behaviours)	`FreeTextLong`	Yes
2.4	Decisions without escalation (examples)	`FreeTextLong`	Yes
2.5	Role profile / objectives / scorecard	`ResponseMode`	No

Step 3: Section 3 — Regular Duties & Cadence

#	Question	Type	Required
3.1	Regular Duties & Cadence list upload	`ResponseMode`	Yes
3.2	Quick-create duties table	`TableGrid`	Fallback
	Columns: Activity / Purpose / Method / Output / Trigger / Frequency / Duration / Target day / R-A		
3.3	Fast fallback: top 10 recurring duties	`FreeTextLong`	Fallback

Step 4: Section 4 — Day-in-the-Life & Time Sinks

#	Question	Type	Required
4.1	Typical day/week (short narrative)	`FreeTextLong`	Yes
4.2	Top 5 recurring interruptions	`FreeTextLong`	Yes
4.3	Non-value-adding time	`FreeTextLong`	Yes
4B.1	Systems used	`MultiSelect`	Yes
	Options: Internal FSM / Client portals / QA tools / Stock system / Mapping-GIS / Time capture / Other		
4B.2	Client portal names (if selected)	`FreeText`	Conditional
4B.3	Where information entered more than once	`FreeTextLong`	Yes
4B.4	Where status mismatches occur	`FreeTextLong`	Yes
4B.5	Portal rules that block closure	`FreeTextLong`	Yes
4B.6	Where you lose most time	`MultiSelect`	Yes
	Options: Logging in / Slow portals / Searching job details / Re-keying / Evidence upload / Error correction / Conflicting instructions		
4B.7	Workaround when systems fail	`FreeTextLong`	Yes
4B.8	Portal guides/screenshots	`ResponseMode`	No
4C.1	What counts as "done" vs "paid"	`FreeTextLong`	Yes
4C.2	Where work gets stuck after physically done	`FreeTextLong`	Yes
4C.3	Backlog tracked? Approximate volume and ageing	`FreeTextLong`	Yes
4C.4	Top 5 reasons jobs remain stuck beyond 48 hours	`FreeTextLong`	Yes
4C.5	Single lever to increase jobs accepted/paid per day	`FreeTextLong`	Yes
4C.6	Throughput uploads	`ResponseMode`	No

Step 5: Section 5 — Operational Reality by Department (CONDITIONAL)

See **Section 3 — Conditional Routing Rules** below for the full routing logic.

The participant sees only the sub-section(s) matching their department selection in Step 1.

5.1 Field Ops — Installations

#	Question	Type	Required
5.1.1	Top 10 install failure reasons	`FreeTextLong`	Yes
5.1.2	Right First Time performance view	`FreeTextLong`	Yes
5.1.3	Early Life Failure view	`FreeTextLong`	Yes
5.1.4	Where installs get blocked	`FreeTextLong`	Yes
5.1.5	Where rework is created	`FreeTextLong`	Yes
5.1.6	Throughput check: done-but-not-paid in installs	`FreeTextLong`	Yes
5.1.7	Install uploads	`ResponseMode`	No

5.2 Field Ops — Service Calls / Repair

#	Question	Type	Required
5.2.1	Top 10 fault types	`FreeTextLong`	Yes
5.2.2	Top 5 repeat fault causes	`FreeTextLong`	Yes
5.2.3	Where fault-to-fix stalls	`FreeTextLong`	Yes
5.2.4	Evidence needed to prove resolution	`FreeTextLong`	Yes
5.2.5	Throughput check: done-but-not-paid in service	`FreeTextLong`	Yes
5.2.6	Service uploads	`ResponseMode`	No

5.3 Field Ops — Pre-Enablement

#	Question	Type	Required
5.3.1	What triggers pre-enablement, common outcomes	`FreeTextLong`	Yes
5.3.2	Biggest "not ready" cause	`FreeTextLong`	Yes
5.3.3	Where it reduces/creates waste	`FreeTextLong`	Yes
5.3.4	Evidence/info for install handover	`FreeTextLong`	Yes
5.3.5	Throughput check: done-but-not-paid	`FreeTextLong`	Yes
5.3.6	Pre-enablement uploads	`ResponseMode`	No

5.4 Field Ops — Enablement Works

#	Question	Type	Required
5.4.1	Top 10 enablement work types	`FreeTextLong`	Yes
5.4.2	Where delays occur most	`FreeTextLong`	Yes
5.4.3	% reactive firefighting vs planned	`FreeTextLong`	Yes
5.4.4	Biggest rework drivers	`FreeTextLong`	Yes
5.4.5	Throughput check: done-but-not-paid	`FreeTextLong`	Yes
5.4.6	Enablement uploads	`ResponseMode`	No

5.5 Dispatch / Scheduling

#	Question	Type	Required
5.5.1	Scheduling approach	`FreeTextLong`	Yes
5.5.2	Top 5 causes of schedule instability	`MultiSelect` + `FreeTextLong` per item	Yes
	Options: Cancellations / Overruns / Priority-SLA escalations / Skills mismatch / Portal-driven changes / FSM-portal sync delays		
5.5.3	Job allocation rules and portal overrides	`FreeTextLong`	Yes
5.5.4	Where dispatch loses most time	`MultiSelect` + `FreeTextLong` per item	Yes
	Options: Missing job data / Portal admin / Re-keying / Chasing updates / Evidence requirements / Preventing disputes		
5.5.5	Most common reason jobs fail to close	`FreeTextLong`	Yes
5.5.6	One system improvement wished for	`FreeTextLong`	Yes
5.5.7	Throughput check: done-but-not-paid backlog	`FreeTextLong`	Yes
5.5.8	Dispatch uploads	`ResponseMode`	No

5.6 Stores / Materials (+ 5.6B + 5.6C)

#	Question	Type	Required
5.6.1	Materials flow description	`FreeTextLong`	Yes
5.6.2	Where losses/waste occur	`FreeTextLong`	Yes
5.6.3	How often stockouts block jobs	`FreeTextLong`	Yes
5.6.4	Root cause of "right kit not on van"	`FreeTextLong`	Yes
5.6.5	Stores uploads	`ResponseMode`	No
5.6B.1	Largest bucket of cash tied up in stock	`FreeTextLong`	Yes
5.6B.2	Overstocked items and why	`FreeTextLong`	Yes
5.6B.3	Understocked items and impact	`FreeTextLong`	Yes
5.6B.4	Stock ageing/slow movers visibility	`FreeTextLong`	Yes
5.6B.5	Returns loop: where it stalls	`FreeTextLong`	Yes
5.6B.6	Biggest driver of stock inaccuracies	`FreeTextLong`	Yes
5.6B.7	What to change to release cash	`FreeTextLong`	Yes
5.6B.8	Inventory uploads	`ResponseMode`	No
5.6C.1	Which clients provide free-issue stock	`FreeTextLong`	Yes
5.6C.2	How client expectations are known	`FreeTextLong`	Yes
5.6C.3	Reconciliation frequency and ownership	`FreeTextLong`	Yes
5.6C.4	Top 5 discrepancy causes	`FreeTextLong`	Yes
5.6C.5	Current alignment state	`SingleSelect`	Yes
	Options: Fully aligned / Minor discrepancies / Major discrepancies / Unknown		
5.6C.6	Discrepancy handling in practice	`FreeTextLong`	Yes
5.6C.7	Free-issue uploads	`ResponseMode`	No

5.7 QA / Quality

#	Question	Type	Required
5.7.1	QA gates description	`FreeTextLong`	Yes
5.7.2	Top 10 recurring defects	`FreeTextLong`	Yes
5.7.3	How findings become corrective action	`FreeTextLong`	Yes
5.7.4	What prevents built-in quality	`FreeTextLong`	Yes
5.7.5	Throughput check: where quality creates done-but-not-paid	`FreeTextLong`	Yes
5.7.6	QA uploads	`ResponseMode`	No

5.8 HSEQ

#	Question	Type	Required
5.8.1	How safety standards are embedded/audited/enforced	`FreeTextLong`	Yes
5.8.2	Stop-work authority: clear and used?	`FreeTextLong`	Yes
5.8.3	Where incidents/near-misses cluster	`FreeTextLong`	Yes
5.8.4	Top 5 HSEQ risks	`FreeTextLong`	Yes
5.8.5	HSEQ uploads	`ResponseMode`	No

5.9 HR

#	Question	Type	Required
5.9.1	Hardest roles to hire and why	`FreeTextLong`	Yes
5.9.2	Onboarding and competence verification	`FreeTextLong`	Yes
5.9.3	Where disciplinaries stall	`FreeTextLong`	Yes
5.9.4	Top 5 people risks	`FreeTextLong`	Yes
5.9.5	HR uploads	`ResponseMode`	No

Step 6: Section 6 — Non-Productive Time

#	Question	Type	Required
6.1	Time loss estimates	`TableGrid`	Yes
	Columns: Category / Estimate		
	Rows: Van-travel / Stores collection / Waiting access / Waiting records / Waiting permits / Rework / Admin / Portal admin / FSM mismatch / Other		
6.2	Time capture uploads	`ResponseMode`	No

Step 7: Section 7 — Cost Leakage

#	Question	Type	Required
7.1	Top 5 avoidable costs in your area	`FreeTextLong`	Yes
7.2	What spend feels normal but is waste	`FreeTextLong`	Yes
7.3	Where penalties/chargebacks/credits arise	`FreeTextLong`	Yes
7.4	Portal/commercial rules creating avoidable cost	`FreeTextLong`	Yes
7.5	Cost leakage uploads	`ResponseMode`	No

Step 8: Section 8 — People Management & Disciplinaries

#	Question	Type	Required
8.1	Open disciplinaries count	`NumberInput`	Yes (if line mgr)
8.2	Where disciplinaries stall	`FreeTextLong`	Yes (if line mgr)
8.3	Top 3 conduct/performance themes	`FreeTextLong`	Yes (if line mgr)
8.4	What support/tools would help	`FreeTextLong`	No

Step 9: Section 9 — Interfaces, Handoffs & Conflicts

#	Question	Type	Required
9.1	3 teams you depend on most + what you need	`FreeTextLong`	Yes
9.2	3 teams that depend on you + what they need	`FreeTextLong`	Yes
9.3	Where handoffs break and consequences	`FreeTextLong`	Yes
9.4	One interface/handoff to fix tomorrow	`FreeTextLong`	Yes
9.5	Where system handoffs break	`FreeTextLong`	Yes

Step 10: Section 10 — Reality Test

#	Question	Type	Required
10.1	Standard items rating (9 items)	`RatingScaleWithEvidence`	Yes
	Rating options (9-point): Exists and is used / Exists but not consistently / Claimed but can't be found / Exists but under-resourced / Implemented but no effect / Tool used for wrong purpose / Conflicting approaches / Works only because key people / Not applicable		
	Items: Planning & scheduling / Dispatch rules / QA gates / Repeat fault loop / Stores control / HSEQ controls / KPI reporting / FSM-portal integration / Portal closure rules		
10.2	Additional items (up to 5)	`RatingScaleWithEvidence`	No

Step 11: Section 11 — Risks, Constraints & What You Would Change

#	Question	Type	Required
11.1	Top 5 constraints blocking performance	`FreeTextLong`	Yes
11.2	Risks you worry about most	`FreeTextLong`	Yes
11.3	One change without extra headcount	`FreeTextLong`	Yes
11.4	One investment you would request	`FreeTextLong`	Yes

Step 12: Section 12 — Evidence Uploads

#	Question	Type	Required
12.1	Evidence upload checklist	`ChecklistUpload`	Encouraged
	Items (14): Local trackers / KPI packs / Rotas / Checklists-SOPs / Meeting notes / Audit results / Exception lists / Portal guides / Job pack requirements / Rejection reasons / Reconciliation trackers / Free-issue statements / Inventory reports / Other		

Step 13: Section 13 — Final Catch-All

#	Question	Type	Required
13.1	What else exists that would help us	`FreeTextLong`	Yes
13.2	Anything we should not misunderstand	`FreeTextLong`	No

4. Wizard Step Flow — 03 Engineer Mini-Audit

Audience: Field engineers (all streams) **Total steps:** 12 (Consent + Sections 1–10) **Estimated time:** 10–12 minutes **Conditional logic:** Section 5 routes by job type (selected in Section 1); multi-skilled engineers answer up to 2 modules

Step 0: Consent

#	Question	Type	Required
0.1	Confidentiality + purpose statement (read-only)	—	Display
0.2	Consent checkbox	`ConsentCheckbox`	Yes — gate

Step 1: Section 1 — About You

#	Question	Type	Required
1.1	Name (optional)	`FreeText`	No
1.2	Role type	`SingleSelect`	Yes
	Options: Install Engineer / Service (Fault) Engineer / Pre-Enablement Engineer / Enablement Works / Multi-skilled		
1.3	Area/region	`FreeText`	Yes
1.4	Employment type	`SingleSelect`	Yes
	Options: Employed / Subcontractor / Other		

Routing trigger: Question 1.2 determines which Section 5 module(s) appear.

Step 2: Section 2 — What You Do & What "Good" Looks Like

#	Question	Type	Required
2.1	Purpose of your role (1 sentence)	`FreeText`	Yes
2.2	What a "good day" looks like (1–2 bullets)	`FreeTextLong`	Yes
2.3	#1 thing that stops a good day	`FreeText`	Yes

Step 3: Section 3 — Non-Productive Time

#	Question	Type	Required
3.1	Time loss estimates	`TableGrid`	Yes
	Columns: Category / Estimate (% or Low-Med-High)		
	Rows: Van-travel / Stores collection / Waiting access / Waiting records / Waiting permits / Rework / Admin / Portal admin / FSM mismatch / Other		
3.2	Top 2 categories that hurt productivity most	`MultiSelectCapped` (max 2)	Yes
3.3	One time sink to remove tomorrow and why	`FreeTextLong`	Yes

Step 4: Section 4 — Systems & Portals Friction

#	Question	Type	Required
4.1	Tools used	`MultiSelect`	Yes
	Options: Internal FSM / Client portals / QA tool / Stock system / Other		
4.2	Client portal names	`FreeText`	Conditional
4.3	Where same info entered twice	`FreeTextLong`	Yes
4.4	Rejection/delay causes at closure	`MultiSelect`	Yes
	Options: Missing photos / Wrong codes / Test results format / Portal timeout / Access-permissions / Conflicting FSM-portal / Other		
4.5	How often system/portal issues delay closure	`SingleSelect`	Yes
	Options: Daily / Weekly / Monthly / Rarely / Never		
4.6	What happens after rejected closure	`FreeTextLong`	Yes
4.7	Screenshot of common rejection	`FileUpload`	No

Step 5: Section 5 — Job Type Modules (CONDITIONAL)

See **Section 3 — Conditional Routing Rules** below.

5A. Installations Module

#	Question	Type	Required
5A.1	Top 5 install failure reasons	`MultiSelectCapped` (max 5)	Yes
	Options: Access issue / Wrong-missing kit / Network not ready / Pre-enablement incomplete / Enablement-civils required / Poor job data / Unrealistic time window / Quality standard unclear / Portal evidence rules / Other		
5A.2	What creates repeat visits most on installs	`FreeTextLong`	Yes
5A.3	One change to improve Right First Time	`FreeTextLong`	Yes
5A.4	Throughput check: done-on-site but not paid frequency and cause	`FreeTextLong`	Yes

5B. Service Calls / Fault Module

#	Question	Type	Required
5B.1	Top 5 fault types	`FreeTextLong`	Yes
5B.2	Biggest repeat fault causes (up to 3)	`MultiSelectCapped` (max 3)	Yes
	Options: Wrong diagnosis / Parts unavailable / Network issue / Evidence constraints / Customer access / Time pressure / Other		
5B.3	Where fault-to-fix stalls most	`FreeTextLong`	Yes
5B.4	What would reduce repeat faults fastest	`FreeTextLong`	Yes
5B.5	Throughput check: done-but-not-paid	`FreeTextLong`	Yes

5C. Pre-Enablement Module

#	Question	Type	Required
5C.1	Most common outcomes (up to 2)	`MultiSelectCapped` (max 2)	Yes
	Options: Ready for install / Blocked – needs enablement / Blocked – access / Reschedule / Job data wrong / Other		
5C.2	Biggest "not ready" cause	`FreeTextLong`	Yes
5C.3	Evidence/info for install handover	`FreeTextLong`	Yes
5C.4	Where pre-enablement reduces/creates waste	`FreeTextLong`	Yes
5C.5	Throughput check: done-but-not-paid	`FreeTextLong`	Yes

5D. Enablement Works Module

#	Question	Type	Required
5D.1	Top 5 enablement job types	`FreeTextLong`	Yes
5D.2	Biggest delay causes (up to 3)	`MultiSelectCapped` (max 3)	Yes
	Options: Permits-TM / Materials-kit / Access / Scope unclear / Coordination / Portal approvals / Other		
5D.3	Biggest rework drivers	`FreeTextLong`	Yes
5D.4	One change to reduce cycle time	`FreeTextLong`	Yes
5D.5	Throughput check: done-but-not-paid	`FreeTextLong`	Yes

Step 6: Section 6 — Materials, Inventory & Free-Issue

#	Question	Type	Required
6.1	How often missing/wrong kit causes delays	`SingleSelect`	Yes
	Options: Daily / Weekly / Monthly / Rarely / Never		
6.2	Where kit problem usually happens (up to 2)	`MultiSelectCapped` (max 2)	Yes
	Options: Van stock not correct / Stores availability / Wrong pick / Returns not processed / Job pack doesn't specify / Other		
6.3	Extra kit carried "just in case"	`FreeTextLong`	Yes
6.4	What happens to unused kit after jobs	`FreeTextLong`	Yes
6.5	Free-issue usage recording and where it goes wrong	`FreeTextLong`	Yes
6.6	Most common reason kit not returned/credited	`FreeTextLong`	Yes

Step 7: Section 7 — Quality & Safety

#	Question	Type	Required
7.1	Standards clear for your jobs?	`SingleSelect`	Yes
	Options: Always / Mostly / Sometimes / Rarely / Never		
7.2	Quality miss causes (up to 3)	`MultiSelectCapped` (max 3)	Yes
	Options: Rushed schedule / Standards unclear / Wrong kit / Poor job data / Portal box-ticking / Lack of training / Other		
7.3	Able to stop work if unsafe?	`SingleSelect`	Yes
	Options: Yes always / Yes usually / Sometimes / No		
7.4	Biggest safety risk seen most often	`FreeTextLong`	Yes

Step 8: Section 8 — Cost Leakage & Chargebacks

#	Question	Type	Required
8.1	Where costs leak most (top 3)	`FreeTextLong`	Yes
8.2	Portal/commercial rules creating waste	`MultiSelect`	Yes
	Options: Evidence requirements / SLA clocks / Closure codes / Repeat visit definitions / No fault found handling / Other		
8.3	One change to reduce chargebacks fastest	`FreeTextLong`	Yes

Step 9: Section 9 — What Would You Change?

#	Question	Type	Required
9.1	One change (no extra headcount) to remove waste	`FreeTextLong`	Yes
9.2	One investment request and why	`FreeTextLong`	Yes
9.3	Anything we should not misunderstand	`FreeTextLong`	No

Step 10: Section 10 — Optional Evidence

#	Question	Type	Required
10.1	Evidence upload	`SingleSelect` + `FileUpload`	No
	Options: Portal rejection screenshot / Job-pack evidence requirements / Example of missing-incorrect data / Nothing to upload		

5. Conditional Routing Rules

5.1 Manager Audit — Section 5 Routing

The department selected in Section 1 (question 1.3) determines which Section 5 sub-section the participant completes:

Department Selection (1.3)	Section 5 Shown	Sub-sections Included
Field Ops – Installations	5.1	5.1 only
Field Ops – Service Calls/Repair	5.2	5.2 only
Field Ops – Pre-Enablement	5.3	5.3 only
Field Ops – Enablement Works	5.4	5.4 only
Dispatch/Scheduling	5.5	5.5 only
Stores/Materials	5.6	5.6 + 5.6B + 5.6C
QA/Quality	5.7	5.7 only
HSEQ	5.8	5.8 only
HR	5.9	5.9 only
Other	None	Skip Section 5 entirely; participant notes their function in 1.4

Special cases: - **Stores/Materials** always gets the extended sub-sections (5.6B Inventory Value, 5.6C Free-Issue). - **"Also relevant to" cross-pollination** — the instrument says "Also relevant to: Dispatch, Stores, QA" on some sections. In the wizard, we do NOT show cross-department sections. Instead, the instrument is designed so that overlapping themes (throughput, materials, quality) appear in the shared sections (4C, 6, 7, 9, 10) that every participant answers. - **"Other" department** — the participant skips Section 5 entirely and provides their departmental context in the catch-all sections (particularly Section 13).

5.2 Engineer Mini-Audit — Section 5 Routing

The role type selected in Section 1 (question 1.2) determines which Section 5 module(s) appear:

Role Selection (1.2)	Modules Shown	Note
Install Engineer	5A only	

Service (Fault) Engineer	5B only	
Pre-Enablement Engineer	5C only	
Enablement Works	5D only	
Multi-skilled	5A + one more (participant selects)	Show secondary module picker

Multi-skilled routing: 1. When "Multi-skilled" is selected, show a secondary question: "Which two work types represent most of your week? (select 2)" 2. Options: Installations / Service / Pre-Enablement / Enablement 3. The two selected modules appear in sequence as Step 5a and Step 5b. 4. If the participant only selects one, show just that module.

5.3 Routing Implementation

```
function getSection5Steps(instrument, profileAnswers) { // Manager Audit — route by
department if (instrument === "02-manager-audit") { const department =
profileAnswers["1.3"]; if (department === "Stores/Materials") { return ["5.6",
"5.6B", "5.6C"]; } else if (department === "Other") { return []; // skip Section 5
entirely } else { return [DEPT_TO_SECTION[department]]; } } // Engineer Mini-Audit —
route by job type if (instrument === "03-engineer-mini-audit") { const roleType =
profileAnswers["1.2"]; if (roleType === "Multi-skilled") { const secondarySelections
= profileAnswers["1.2b"]; return secondarySelections.map(s => "5" + s); // e.g.
["5A", "5B"] } else { return [ROLE_TO_MODULE[roleType]]; } } return []; // Company
Audit has no Section 5 routing }
```

6. Data Model

6.1 Entity Relationship

```
Engagement 1■■* Participant Engagement 1■■* InstrumentConfig Participant 1■■*
Submission Submission 1■■* SectionDraft Submission 1■■* Upload SectionDraft 1■■*
Upload
```

6.2 Entities

Engagement

```
{ "id": "uuid", "client_name": "Nano Fibre UK Limited", "client_legal_name": "NANO FIBRE UK
LIMITED", "supplier_legal_name": "TUMAI MANAGEMENT LTD", "slug": "nano-fibre-uk", "subdomain":
"nano", "status": "active", "audit_start_date": "2026-03-10", "interview_start_date": "2026-03-16",
"contact_name": "Greg Kurnikov", "contact_email": "greg@limitlessmodus.com", "created_at":
"2026-03-02T00:00:00Z", "azure_tenant_ids": ["nano-fibre-tenant-id"], "allowed_email_domains":
["nanois.co.uk", "nanofibre.co.uk"] }
```

Participant

```
{ "id": "uuid", "engagement_id": "uuid", "name": "Ashley Wass", "email": "ashley.w@nanofibre.co.uk",
"role_title": "Head of Operations", "department": "Operations", "participant_group":
"heads-of-department", "instruments_assigned": ["02-manager-audit"], "section5_preset": "5.1",
"is_anonymous": false, "joint_submission_group": null, "notes": "Internal coordinator",
"created_at": "2026-03-02T00:00:00Z" }
```

Submission

```
{ "id": "uuid", "participant_id": "uuid", "instrument_id": "01-company-audit", "status": "draft",
"consent_given": true, "consent_timestamp": "2026-03-10T09:15:00Z", "profile_answers": { "1.3":
"Field Ops – Installations" }, "section5_routing": ["5.1"], "progress_percent": 45,
"sections_completed": ["consent", "1", "2"], "sections_total": 15, "first_opened_at":
"2026-03-10T09:00:00Z", "last_saved_at": "2026-03-10T14:30:00Z", "submitted_at": null, "locked":
false }
```

Status values: `not_started` → `draft` → `submitted` → `locked`

SectionDraft

```
{ "id": "uuid", "submission_id": "uuid", "section_key": "A", "answers": { "A1.1": "We build and
maintain broadband networks...", "A1.2": "Safety first, quality always, customer focus...", "A2.1":
{ "mode": "upload", "file_id": "uuid" }, "A3.1": { "mode": "not_available", "note": "Will cover in
interview" } }, "is_complete": true, "last_saved_at": "2026-03-10T11:22:00Z" }
```

Upload

```
{ "id": "uuid", "submission_id": "uuid", "section_key": "B", "question_key": "B2.2",
"original_filename": "Installs-Regular-Duties.xlsx", "stored_path":
"engagements/nano-fibre-uk/submissions/uuid/B/B2.2/file.xlsx", "mime_type":
"application/vnd.openxmlformats-officedocument.spreadsheetml.sheet", "size_bytes": 45200,
"uploaded_at": "2026-03-10T12:00:00Z", "metadata": { "what_it_proves": "Regular duties cadence for
Installations team" } }
```

6.3 Submission Lifecycle State Machine

```
■■■■■■■■■■■■■■■■ ■ not_started ■ Participant has not opened the audit ■■■■■■■■■■■■■■■■ ■
Consent given v ■■■■■■■■■■■■■■■■ ■ draft ■ Sections being completed, auto-saved
■■■■■■■■■■■■■■■■ ■ All required sections answered + "Submit" clicked v ■■■■■■■■■■■■■■■■ ■
submitted ■ Read-only for participant, visible to admin ■■■■■■■■■■■■■■■■ ■ Admin locks (or
auto-lock after review period) v ■■■■■■■■■■■■■■■■ ■ locked ■ Immutable, ready for synthesis
■■■■■■■■■■■■■■■■ Reverse transitions: - submitted → draft : Admin unlocks for amendments
(logged) - locked → submitted : Admin unlocks (exceptional, logged)
```

6.4 Joint Submission (Directors)

When two directors submit jointly (e.g., Wilhelm + Dirk): - Both Participant records reference the same `joint_submission_group` ID. - A single Submission record is created, owned by the group. - Both participants can edit the same submission (last-write-wins per section, with section-level locking). - Both names appear on the submission.

7. Authentication and Authorisation

7.1 Azure Entra ID (SSO)

Aspect	Detail
Provider	Microsoft Entra ID (Azure AD)
Flow	Authorization Code Flow with PKCE (Nuxt frontend)
App Registration	Multi-tenant app, restricted to configured tenant IDs + email domains
Redirect URI	`https://nano.limitlessmodus.com/auth/callback`
Scopes	`openid`, `profile`, `email`
Token storage	HTTP-only secure cookie (access token) + refresh token in server session

7.2 Participant Matching

After SSO login, the portal matches the authenticated email to a Participant record:

1. User signs in via Microsoft Entra ID.
2. Portal receives `id_token` with `email` claim.
3. Backend looks up `participant.email` matching the authenticated email (case-insensitive).
4. If match found: show participant dashboard with their assigned audit(s).
5. If no match: show "You are not registered for an active engagement. Contact your coordinator."

7.3 Roles

Role	Access	How Assigned
`participant`	Own submission(s) only: view, edit, submit	Auto-matched by email
`admin`	All engagements, all submissions, export, unlock	Manual flag on Participant record
`viewer`	Read-only access to submissions + exports	Manual flag (for Greg to review)

7.4 Email Domain Restrictions

Each engagement defines `allowed_email_domains`. The SSO callback validates that the authenticated email domain is in the allowed list before granting access. Admin accounts (Tumai staff) bypass domain restrictions.

8. Participant Journey

8.1 Invitation Flow

1. **Admin creates engagement** in the portal with participant list.
2. **Portal generates personalised invitation emails** (or admin sends manually) containing: - Portal URL: `https://nano.limitlessmodus.com` - Brief explanation of the audit process - Time estimate for their specific instrument - Deadline (audit close date) - Support contact
3. **Participant clicks link**, is redirected to Microsoft SSO login.
4. **After login**, portal matches email → shows dashboard.

8.2 Participant Dashboard

The dashboard shows:

Element	Detail
Welcome banner	"Welcome, Ashley. Your Invincibility Blueprint audit is ready."
Assigned audit card	Instrument name, estimated time, deadline, current status
Progress bar	Sections completed / total sections
"Continue" button	Resumes from last incomplete section
Section navigator	Clickable list of all sections with status (not started / in progress / complete)
Support link	Contact coordinator

8.3 Wizard Experience

1. **Step 0 (Consent)** — must be completed first. Cannot proceed without consent.
2. **Steps 1..N** — one section per step. Can navigate forward/back. Auto-saves on navigate away.
3. **Progress indicator** — top of page shows: step X of Y, section name, progress bar.
4. **Section status** — each section shows: "Not started" / "In progress" / "Complete" in the sidebar navigator.
5. **ResponseMode** — wherever evidence is requested, the four-mode radio group appears. Selecting "Upload" expands a file drop zone. Selecting "Will send later" expands an ETA text field.
6. **Review step** — after all sections, a review screen shows all answers summary with "Edit" links per section.
7. **Submit** — locks the submission. Confirmation page with timestamp and "Your submission has been received."

8.4 Mobile Experience (Engineer Mini-Audit)

The Engineer Mini-Audit is designed for 10–12 minutes on a phone: - Single-column layout - Large touch targets for radio/checkbox inputs - Minimal typing (most questions are select-based) - File upload simplified (camera capture for screenshots) - Progress saved automatically — can close and resume

9. Admin Dashboard Requirements

9.1 Engagement Overview

Element	Detail
Engagement header	Client name, dates, status
Submission tracker table	27 rows (one per participant)
Columns	Name / Role / Department / Instrument / Status / % Complete / Last Activity / Actions
Status badges	Not started (grey) / In progress (amber) / Submitted (green) / Locked (blue)
Filters	By instrument type, by status, by department

9.2 Submission Viewer

- Read-only view of a submitted audit
- Section-by-section navigation
- Inline display of text answers
- File download links for uploads
- Export to JSON / Markdown for AI synthesis

9.3 Export

Format	Purpose
JSON	Structured data for AI synthesis pipeline
Markdown	Human-readable per-submission report
CSV	Tabular summary of all submissions (one row per participant, columns per key question)
ZIP	All uploads for a submission or engagement

9.4 Admin Actions

- **Send reminder** — trigger email to participants who haven't started or are in progress
- **Unlock submission** — revert submitted/locked back to draft (logged with reason)
- **Close engagement** — prevent new submissions, lock all drafts
- **Export all** — bulk download all submissions + uploads

10. File Upload and Evidence Management

10.1 Upload Rules

Aspect	Rule
Max file size	50 MB per file
Max files per question	5
Allowed types	PDF, DOCX, XLSX, PPTX, CSV, PNG, JPG, JPEG, GIF, BMP, TXT, ZIP
Storage	Azure Blob Storage (engagement-scoped containers)
Naming	`{engagement}/{submission_id}/{section}/{question}/{original_filename`
Virus scan	On upload (block if detected)
Upload guidance	Shown above every FileUpload widget: minimise personal identifiers, no special category data, redact if needed

10.2 ResponseMode Widget

Every evidence request uses the ResponseMode pattern:

```
■■■
```

11. Auto-save, Progress Tracking, and Submission Lifecycle

11.1 Auto-save

- Every field change is auto-saved after a 2-second debounce.
- Section navigation triggers an immediate save of the current section.
- Browser tab close / navigation away triggers a save (via `beforeunload`).
- Save indicator in the header: "Saved" / "Saving..." / "Save failed (retry)".
- Conflict resolution: last-write-wins per field. No concurrent editing expected (one participant per submission, except joint director submissions).

11.2 Progress Tracking

- **Section-level:** each section marked as Not started / In progress / Complete.
- **Complete** = all required fields in the section have a value (or ResponseMode selection).
- **Overall progress** = completed sections / total sections (percentage).
- **Progress bar** shown on dashboard and in wizard header.

11.3 Submission Rules

- Participant can submit when: consent given AND at least 80% of required sections are complete.
- Incomplete optional sections do not block submission.
- On submit: all section drafts are frozen, status changes to `submitted`, timestamp recorded.
- Post-submit: participant sees read-only view of their submission with a "Request amendment" button (sends email to admin).

12. API Contract Outline

12.1 Authentication

Method	Endpoint	Description
GET	`/auth/login`	Redirect to Azure SSO
GET	`/auth/callback`	Handle SSO callback, create session
POST	`/auth/logout`	Destroy session
GET	`/auth/me`	Current user info + participant match

12.2 Participant API

Method	Endpoint	Description
GET	`/api/participant/dashboard`	Assigned instruments, submission status
GET	`/api/participant/submission/{id}`	Full submission with all section drafts
PUT	`/api/participant/submission/{id}/section/{key}`	Save section draft (auto-save)
POST	`/api/participant/submission/{id}/submit`	Final submission
POST	`/api/participant/submission/{id}/upload`	Upload file to section/question
DELETE	`/api/participant/submission/{id}/upload/{uploadId}`	Remove uploaded file

12.3 Admin API

Method	Endpoint	Description
GET	`/api/admin/engagements`	List all engagements
GET	`/api/admin/engagements/{id}`	Engagement detail + participant list
GET	`/api/admin/engagements/{id}/submissions`	All submissions with status
GET	`/api/admin/submissions/{id}`	Full submission read-only
POST	`/api/admin/submissions/{id}/unlock`	Revert to draft
POST	`/api/admin/submissions/{id}/lock`	Lock submission
GET	`/api/admin/engagements/{id}/export`	Export all submissions (JSON/CSV/ZIP)
POST	`/api/admin/engagements/{id}/remind`	Send reminder emails

12.4 Instrument Definition API

Method	Endpoint	Description
GET	`/api/instruments`	List all instrument definitions
GET	`/api/instruments/{id}`	Full instrument definition (steps, questions, routing)

Instrument definitions are stored as structured JSON matching the question inventory in Sections 2–4 of this spec. The frontend renders the wizard dynamically from these definitions.

Appendix A — Nano Fibre UK Engagement Configuration

A.1 Engagement Record

Field	Value
Client	Nano Fibre UK Limited
Slug	`nano-fibre-uk`
Subdomain	`nano` (→ `nano.limitlessmodus.com`)
Audit start	10 March 2026
Interview start	16 March 2026
Coordinator	Ashley Wass (`ashley.w@nanofibre.co.uk`)
Consultant	Greg Kurnikov
Allowed email domains	`nanois.co.uk`, `nanofibre.co.uk`

A.2 Participant-to-Instrument Mapping (27 participants)

Directors — Company Audit (01) + Director Interview (04)

#	Name	Email	Instrument	Section 5 Preset	Joint Group
1	Wilhelm Strumpher	wilhelm@nanois.co.uk	01-company-audit	N/A	directors-joint
2	Dirk Mostert	dirk@nanois.co.uk	01-company-audit	N/A	directors-joint

Heads of Department — Manager Audit (02)

#	Name	Email	Instrument	Section 5 Preset
3	Andre Steyn	andre@nanofibre.co.uk	02-manager-audit	Other (CFO)
4	Anna Hanekom	anna@nanofibre.co.uk	02-manager-audit	Other (Finance & Payroll)
5	Ashley Wass	ashley.w@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
6	Lisa Woods	lisa.w@nanois.co.uk	02-manager-audit	5.9 (HR)
7	Callum Krzysik	callum.k@nanofibre.co.uk	02-manager-audit	5.7 (QA/Quality)
8	Andy Green	andrew.g@nanofibre.co.uk	02-manager-audit	5.8 (HSEQ)
9	Roland Dumas	roland.d@nanofibre.co.uk	02-manager-audit	Other (Fleet)
10	Stephan Theron	stephan@nanois.co.uk	02-manager-audit	Other (Development)
11	Gavin Estehuizen	gavin.e@nanofibre.co.uk	02-manager-audit	5.6 (Stores/Materials)

Installation Managers — Manager Audit (02)

#	Name	Email	Instrument	Section 5 Preset
12	Adam Winning	adam.w@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
13	Damon Reid	damon.r@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
14	Learnmore Gonda	learnmore.g@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
15	Aaron Balmer	aaron.b@nanofibre.co.uk	02-manager-audit	5.1 (Installations)

Pre-enablement & Remediations Managers — Manager Audit (02)

#	Name	Email	Instrument	Section 5 Preset
16	Kevin Du Plessis	kevin.dp@nanofibre.co.uk	02-manager-audit	5.3 (Pre-Enablement)
17	Jordan Middleton	jordan.m@nanofibre.co.uk	02-manager-audit	5.3 (Pre-Enablement)
18	Ruben Swart	ruben.s@nanofibre.co.uk	02-manager-audit	5.3 (Pre-Enablement)
19	Hanlo Naude	hanlo.n@nanofibre.co.uk	02-manager-audit	5.3 (Pre-Enablement)

Supervisors — Manager Audit (02)

#	Name	Email	Instrument	Section 5 Preset
20	Jack Lunt	jack.l@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
21	Calvin Chimedza	calvin.c@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
22	Arno De Greeff	arno.dg@nanofibre.co.uk	02-manager-audit	5.1 (Installations)
23	Micheal Hibbert	hibbert.m@nanofibre.co.uk	02-manager-audit	5.1 (Installations)

Engineers — Engineer Mini-Audit (03)

#	Name	Email	Instrument	Anonymous	Note
24	JP Rowan	john-paul.r@nanofibre.co.uk	03-engineer-mini-audit	Optional	Interview optional
25	Nicholas Mensforth	nicholas.m@nanofibre.co.uk	03-engineer-mini-audit	Optional	Interview optional
26	Mateusz Biernacki	Mateusz.b@nanofibre.co.uk	03-engineer-mini-audit	Optional	Email has capital M
27	Mathew Morley	matt.m@nanofibre.co.uk	03-engineer-mini-audit	Optional	Interview optional

A.3 Section 5 Preset Notes

Some participants have departments that don't map directly to a Section 5 sub-section (CFO, Finance & Payroll, Fleet, Development). These participants get `section5_preset = "Other"` and skip Section 5, answering only the universal sections (1–4, 6–13). The "Other" field in Section 1 (question 1.4) captures their functional area.

Ashley Wass (Head of Operations) is mapped to 5.1 (Installations) as her primary operational domain, consistent with her role overseeing field delivery.

Supervisors are all mapped to 5.1 (Installations) as they supervise field installation teams.

A.4 Open Questions from Onboarding

1. **Andre Steyn (CFO)** — should he also contribute to the Company Audit alongside directors? If yes, add instrument `01-company-audit` to his record.
2. **Supervisors** — currently assigned Manager Audit. If Greg prefers a lighter touch, switch to Engineer Mini-Audit.
3. **Email case sensitivity** — Mateusz.b@nanofibre.co.uk has capital M. Portal email matching must be case-insensitive.

Part of the Invincibility Blueprint Toolkit — Limitless Modus.